

Appendix:

This appendix gives step by step instruction using the tools in ArcGIS Pro (version 3.6.0) to complete the study reported. All datasets were made available in a folder on the computer, data-raw, which was split into 4 sub folders : a62, wind-speed, national-parks, penrith. At each stage used in the analysis, new map data was stored in a single file geodatabase called exam.gdb. Unless stated otherwise, all Environment settings (cell size and extent) should be matched to dem.tif. Analysis was performed across the entire Penrith area before being clipped into study area a62.

Start of Appendix

1. Project Setup and Loading Wind Speed Data

- Open ArcGIS Pro, create new project with file geodatabase (exam.gdb).
- Add Data: dem.tif
- Data Management Tools | Raster | Raster Datasets | Copy Raster
 - Input: MeanWindSpeed_Apr_1981-2010_LTA.asc
 - Output: windspeed_apr
 - Environments → Output Coordinate System: dem.tif

2. Downscaling Wind Speed to 30 m Resolution

- Conversion Tools | From Raster | Raster to Point
 - Input: windspeed_apr | Output: points5km
- Conversion Tools | To Raster | Feature to Raster
 - Input: points5km | Value field: pointid | Cell size: 5000 | Output: zones5km
- Spatial Analyst Tools | Zonal | Zonal Statistics
 - Zone raster: zones5km | Value raster: dem.tif | Statistic: MEAN | Output: demmean
- Spatial Analyst Tools | Map Algebra | Raster Calculator
 - $\text{Power}(((\text{"dem.tif"} + 80) / (\text{"demmean"} + 10)), 0.143) * (\text{"windspeed_apr"} * 0.51444)$
 - Output: windspeedms

3. Estimating Power and Energy Generation

- Spatial Analyst Tools | Map Algebra | Raster Calculator
 - $\text{Con}(\text{"windspeedms"} > 3, (230 * \text{"windspeedms"} - 690), 0)$
 - Output: powerKW
- Spatial Analyst Tools | Map Algebra | Raster Calculator
 - $\text{"powerKW"} * 720$
 - Output: energy_apr

4. Screening Out Unsuitable Areas

- Spatial Analyst Tools | Surface | Slope
 - Input: dem.tif | Output measurement: DEGREE | Output: gradient

- Spatial Analyst Tools | Distance | Legacy | Euclidean Distance
 - Input: roads.shp | Output: roadsdist
- Spatial Analyst Tools | Map Algebra | Raster Calculator
 - Con(("landcov.tif" == 22) | ("landcov.tif" == 23), 1)
 - Output: urban
- Spatial Analyst Tools | Distance | Legacy | Euclidean Distance
 - Input: urban | Output: urbdist
- Spatial Analyst Tools | Distance | Legacy | Euclidean Distance
 - Input: settlements.shp | Output: settldist
- Conversion Tools | To Raster | Feature to Raster
 - Input: national_parks.shp | Value Field : FID_1 | Output: natpark_r
- Spatial Analyst Tools | Map Algebra | Raster Calculator — combine all constraints:
 - Con(("dem.tif" < 300) & ("gradient" < 10) & ("roadsdist" <= 3000) & ("urbdist" > 2000) & ("settdist" > 500) & IsNull("natpark_r"), 1)
 - Output: suitablemask
- Spatial Analyst Tools | Map Algebra | Raster Calculator
 - Con("suitablemask" == 1, "energy_apr")
 - Output: energy_suitable

5. Identifying Top Candidate Sites

- Spatial Analyst Tools | Reclass | Slice
 - Input: energy_suitable | Output zones: 200 | Method: EQUAL AREA | Output: energy_slice
- Spatial Analyst Tools | Map Algebra | Raster Calculator
 - Con("energy_slice" == 200, 1)
 - Output: topsites
- Conversion Tools | From Raster | Raster to Polygon
 - Input: topsites | Output: topsites_poly
- Spatial Analyst Tools | Zonal | Zonal Statistics as Table
 - Zone data: topsites_poly | Zone field: OBJECTID | Value raster: energy_suitable
 - Statistic: MEAN | Output: sites_ranked
- Analysis Tools | Extract | Select
 - Input: topsites_poly | Output: top3_sites
 - Expression: OBJECTID = 9 OR OBJECTID = 4 OR OBJECTID = 3

6. Viewshed Analysis

- Data Management Tools | Features | Feature To Line
 - Input: top3_sites | Output: top3_lines
- Spatial Analyst Tools | Surface | Visibility
 - Input raster: dem.tif | Observer features: top3_lines | Output: viewshed
 - Observer parameters → Outer Radius: 20000
 - Environments → Cell size: 30 m
- Spatial Analyst Tools | Map Algebra | Raster Calculator
 - Con("viewshed" > 0, 1, 0)
 - Output: view_binary

7. Visibility per LSOA and Population Linkage

- Spatial Analyst Tools | Zonal | Zonal Statistics as Table
 - Zone data: Isoa.shp | Zone field: code | Value raster: view_binary
 - Statistic: ALL | Output: vis_table
- Add Data: census2011.csv
- Right-click Isoa → Joins and Relates → Add Join
 - Input Join Field: code | Join Table: census2011.csv | Join Table Field: geography code
- Right-click Isoa → Joins and Relates → Add Join
 - Input Join Field: code | Join Table: vis_table | Join Table Field: ZONE_CODE
- Right-click Isoa → Data → Export Features | Output: Isoa_vis_pop
- Right-click Isoa_vis_pop → Data → Export Table | Save as .csv to project folder

8. Clipping Results to Study Area a62

- Spatial Analyst Tools | Extraction | Extract by Mask (repeat for all raster outputs)
 - Input raster: (e.g. energy_suitable) | Mask: a62.shp | Output: (e.g. energy_a62)
- Analysis Tools | Extract | Clip (repeat for all vector outputs)
 - Input: (e.g. top3_sites) | Clip Features: a62.shp | Output: (e.g. top3_a62)

End of Appendix